

Question 1

Hamburgers Price = \$2			
Quantity	Total Utility	Marginal Utility	Marginal Utility Per Dollar
0	0	0	0
1	12	12	6
2	20	8	4
3	26	6	3
4	30	4	2
5	32	2	1

Soft Pretzels Price = \$3			
Quantity	Total Utility	Marginal Utility	Marginal Utility Per Dollar
0	0	0	0
1	24	24	8
2	42	18	6
3	54	12	4
4	63	9	3
5	69	6	2

If the budget given to Sam at the local county fair for food is \$18, what would be the combination of hamburgers and soft pretzels that would maximize his utility?

Answer

3 hamburgers 4 soft pretzels

Explanation

1. He would first buy ONE soft pretzel because the MU/P of 1 soft pretzel is 8 and the MU/P of one hamburger is 6. He will be left over with \$15.
2. Next, he will buy a SECOND soft pretzel (MU/P = 6) and a FIRST hamburger (MU/P = 6). He will have \$10 at this point.
3. Then, he will buy a SECOND hamburger (MU/P = 4) and a THIRD soft pretzel (MU/P = 4). He will end up with \$5 after this.
4. Finally, to use the last of his budget, he would buy his THIRD hamburger (MU/P = 3) and the FOURTH soft pretzel (MU/P = 3)

The ideal combination would be 3 hamburgers and 4 soft pretzels. MU/P of 3 hamburgers = MU/P of 4 soft pretzels (3 = 3), while still staying in budget

Question 2

	Popcorn Buckets	Large Sodas
Price per Unit	\$5	\$3
Marginal Utility	25	18

The table below shows the per-unit prices and marginal utility for the last unit of popcorn buckets and large sodas that Donna purchased. Donna spent all of her allocated budget on buckets of popcorn and large sodas at the movies. To maximize her utility, Donna should have purchased:

- (A) more buckets of popcorn and fewer large sodas
- (B) fewer buckets of popcorn and more large sodas
- (C) fewer of both goods
- (D) equal amounts of both goods
- (E) more of both goods

Answer

Choice B

Explanation

Donna should purchase more large sodas and less popcorn buckets because the MU/P of large sodas is greater than the MU/P of buckets of popcorn (MU/P of large sodas is $18/3$, which is 6, and the MU/P of buckets of popcorn is $25/5$, which is 5). Since $6 > 5$, large soda has a greater utility-per-dollar value than popcorn buckets.

The rule of thumb is: if the MU/P for the two goods are not equal, then you buy more of the higher value good and less of the lower value good.